

Figure 1. Epigenetic Clock Validation and Performance

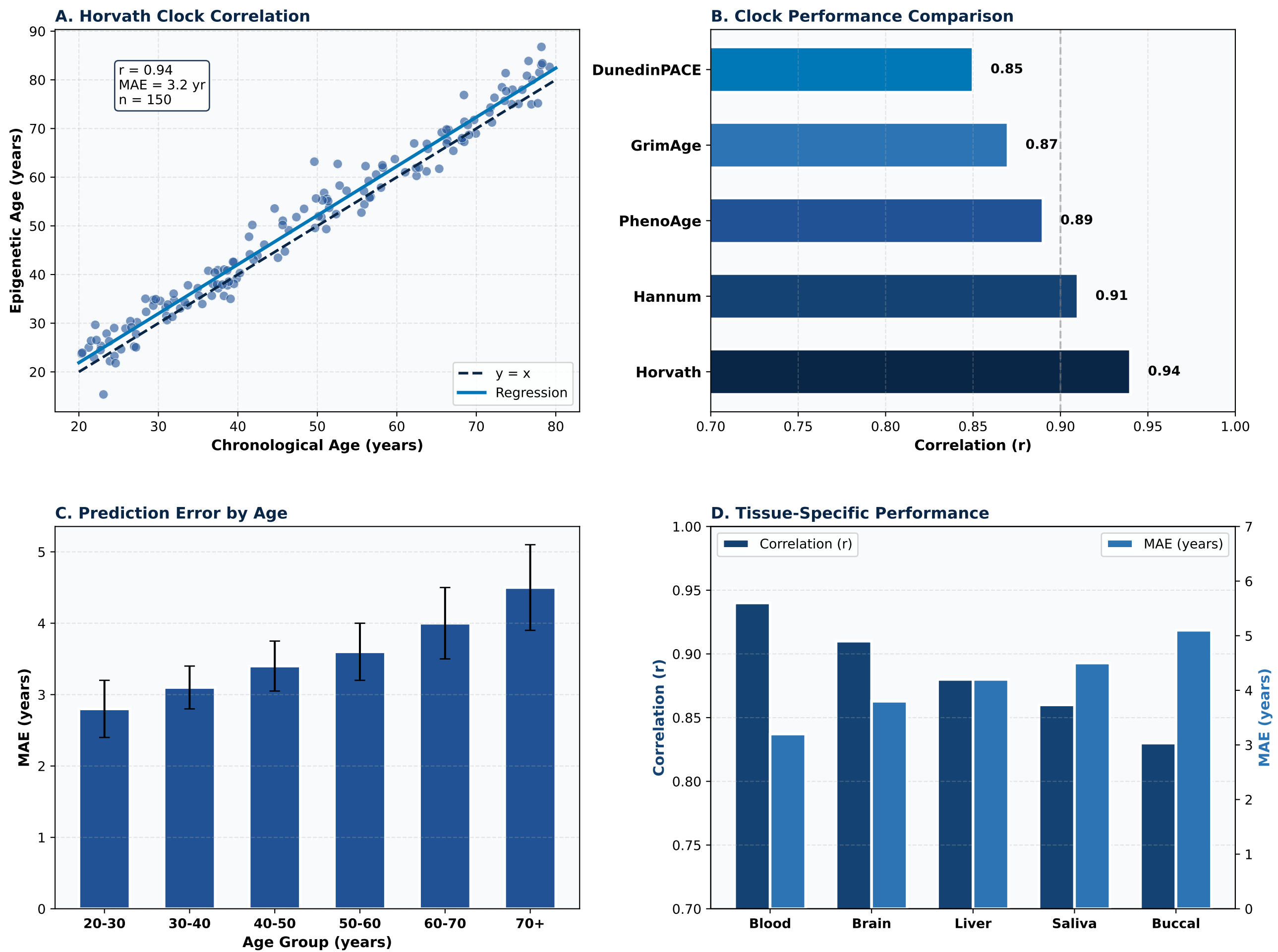
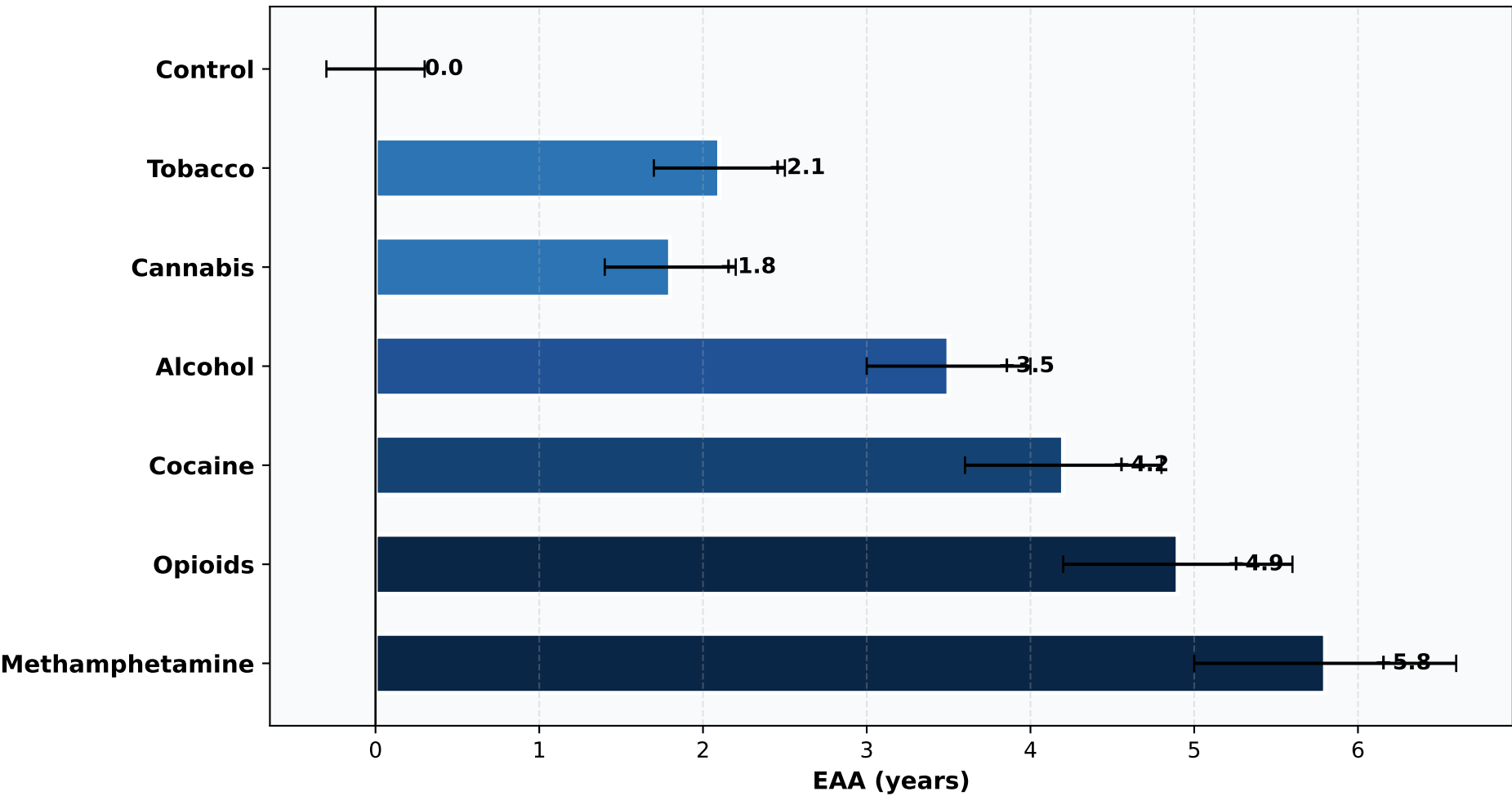
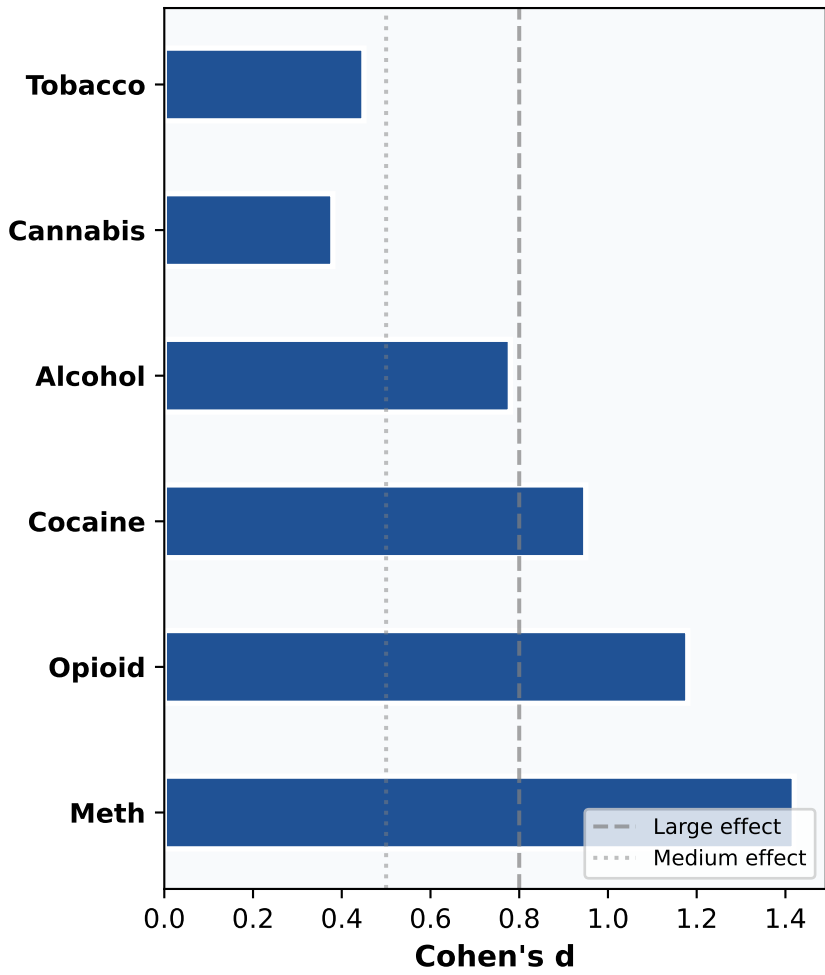


Figure 2. Substance-Specific Epigenetic Age Acceleration

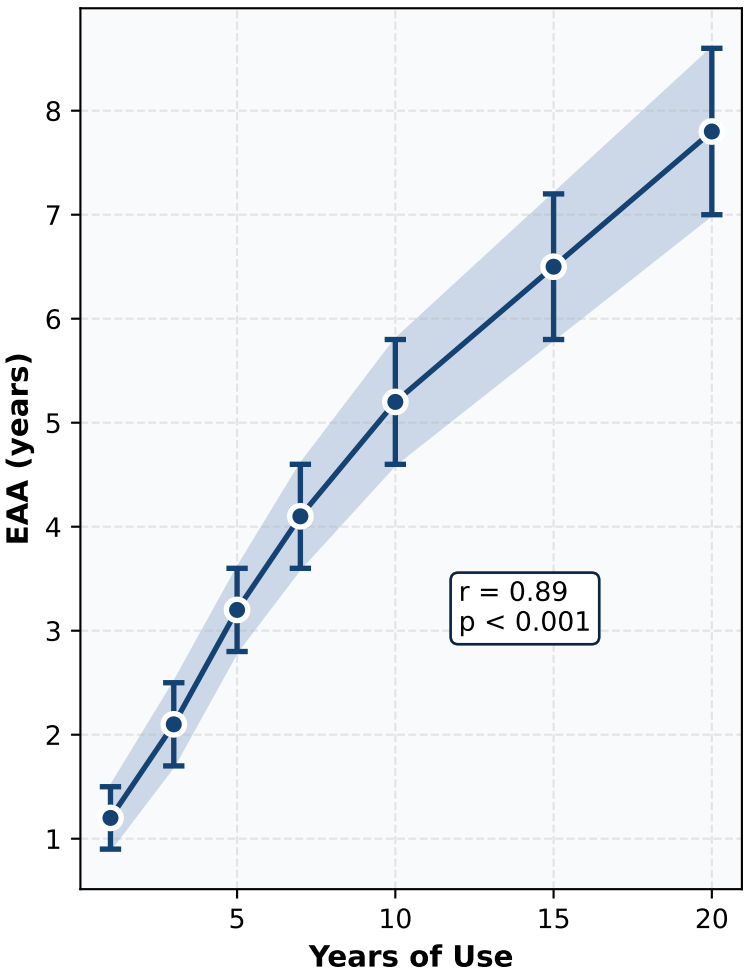
A. Epigenetic Age Acceleration by Substance



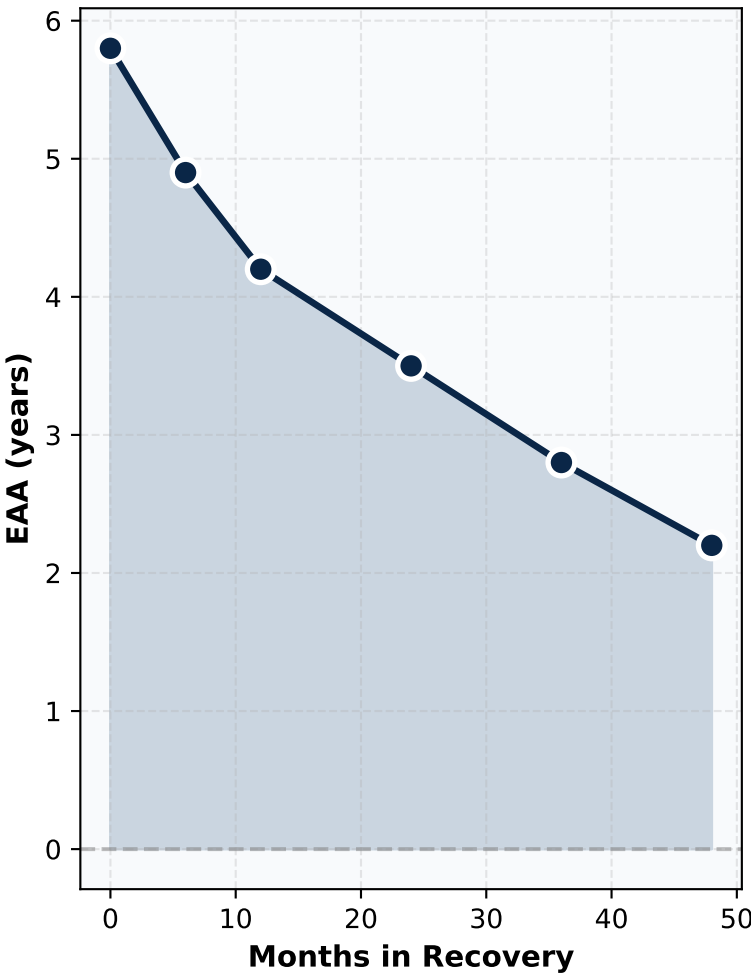
B. Effect Sizes



C. Dose-Response (Methamphetamine)



D. EAA During Recovery



E. Polysubstance Effect

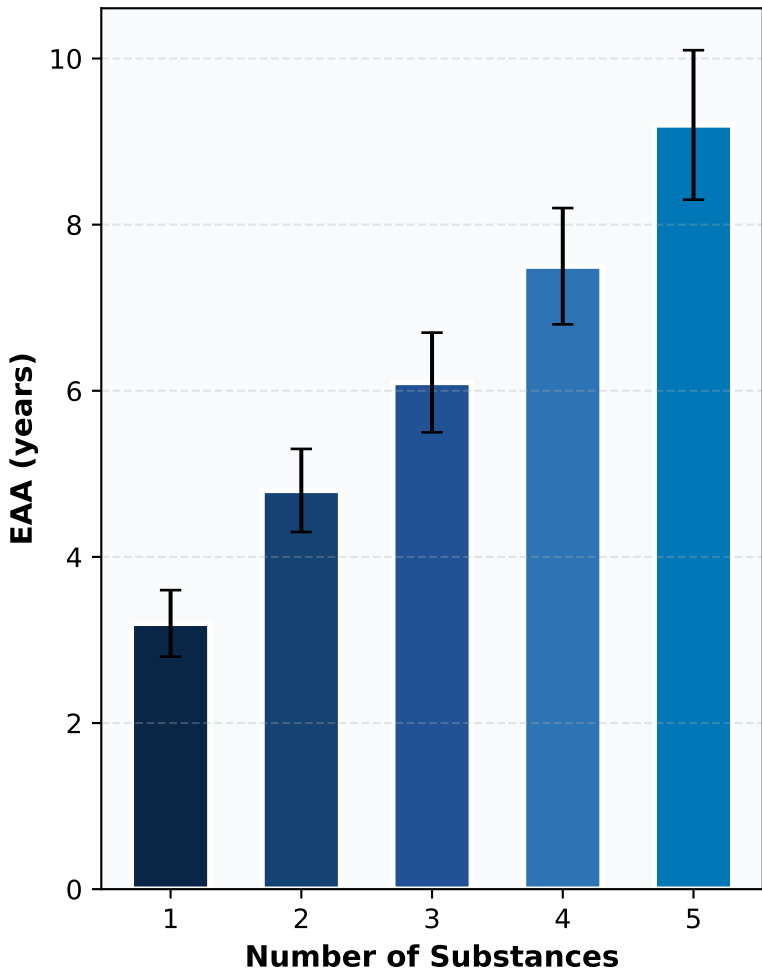
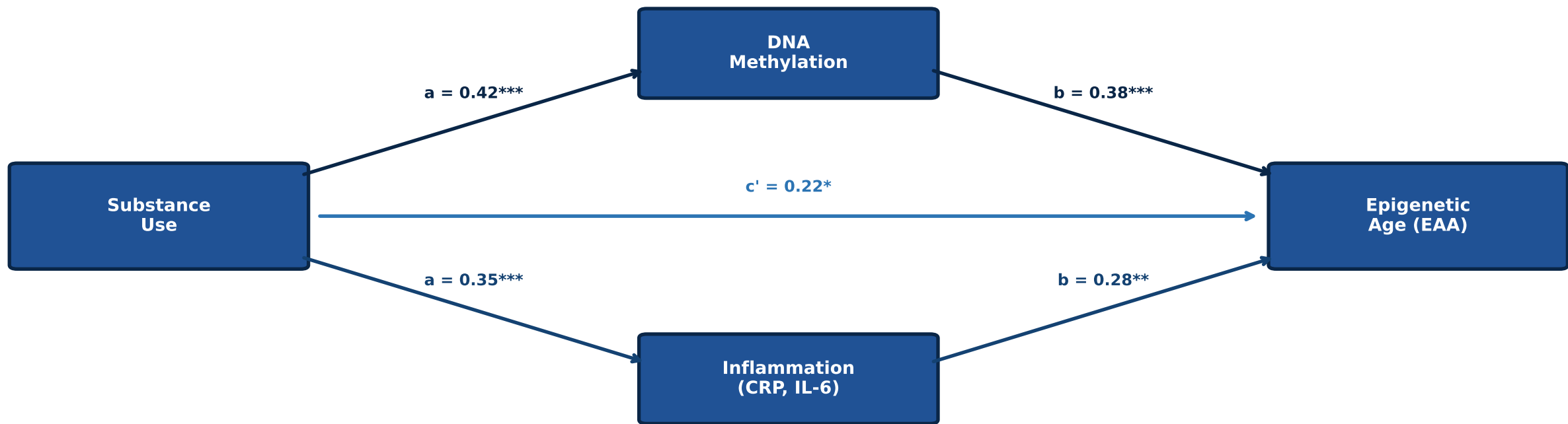
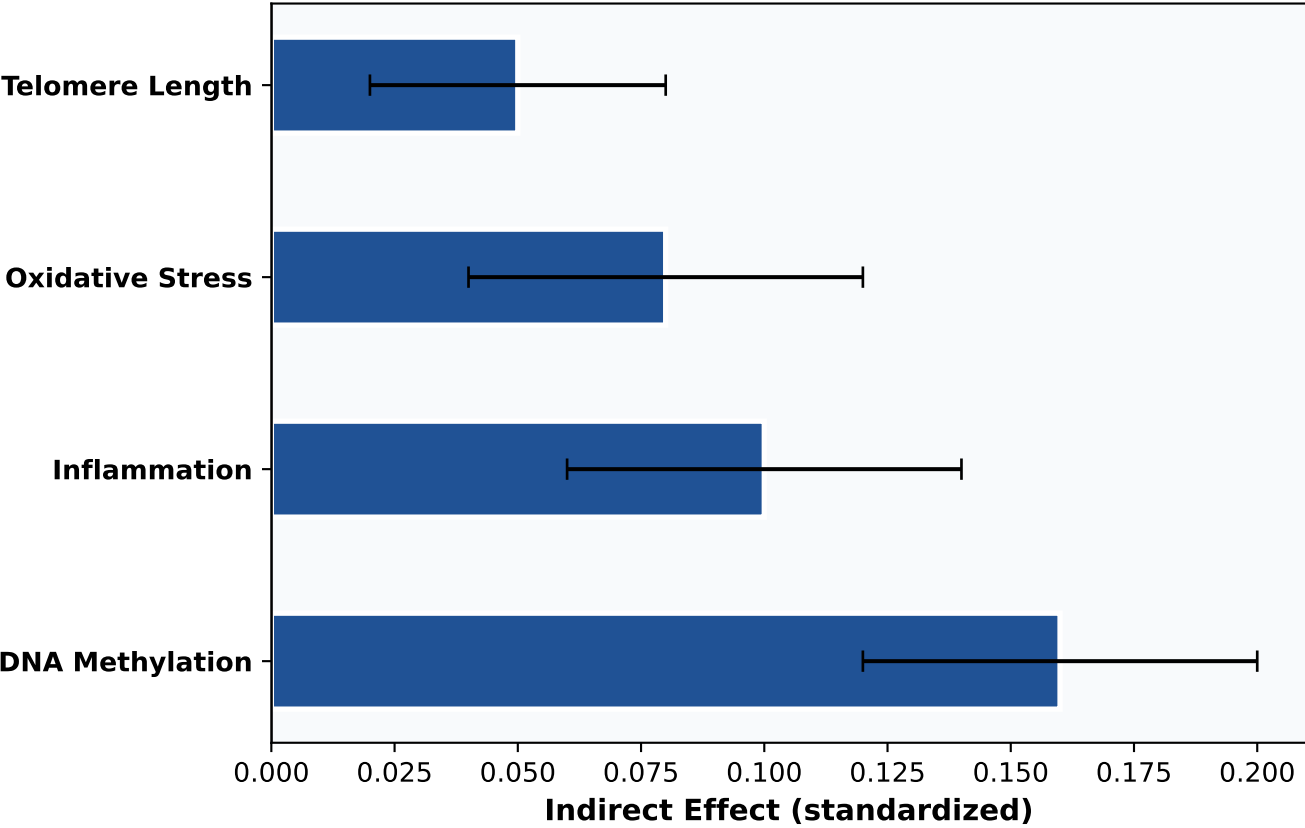


Figure 3. Mediation Analysis: Mechanisms of Epigenetic Aging

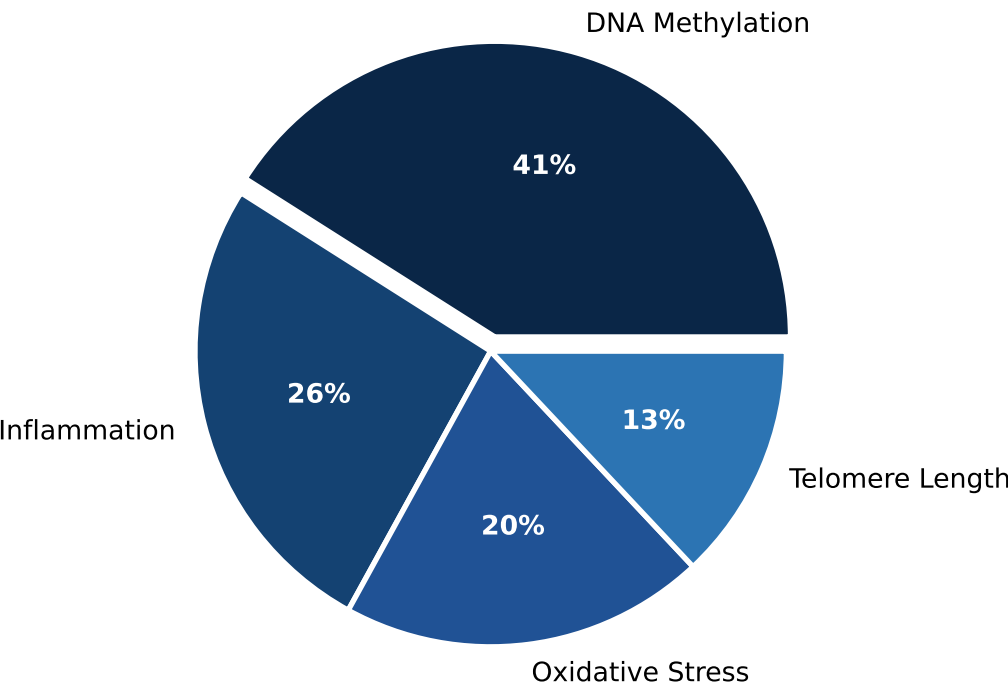
A. Mediation Path Model



B. Mediation Effects



C. Proportion of Total Effect Mediated



Bootstrap 10,000 iterations | ***p<0.001, **p<0.01, *p<0.05

Figure 4. Sample Characteristics

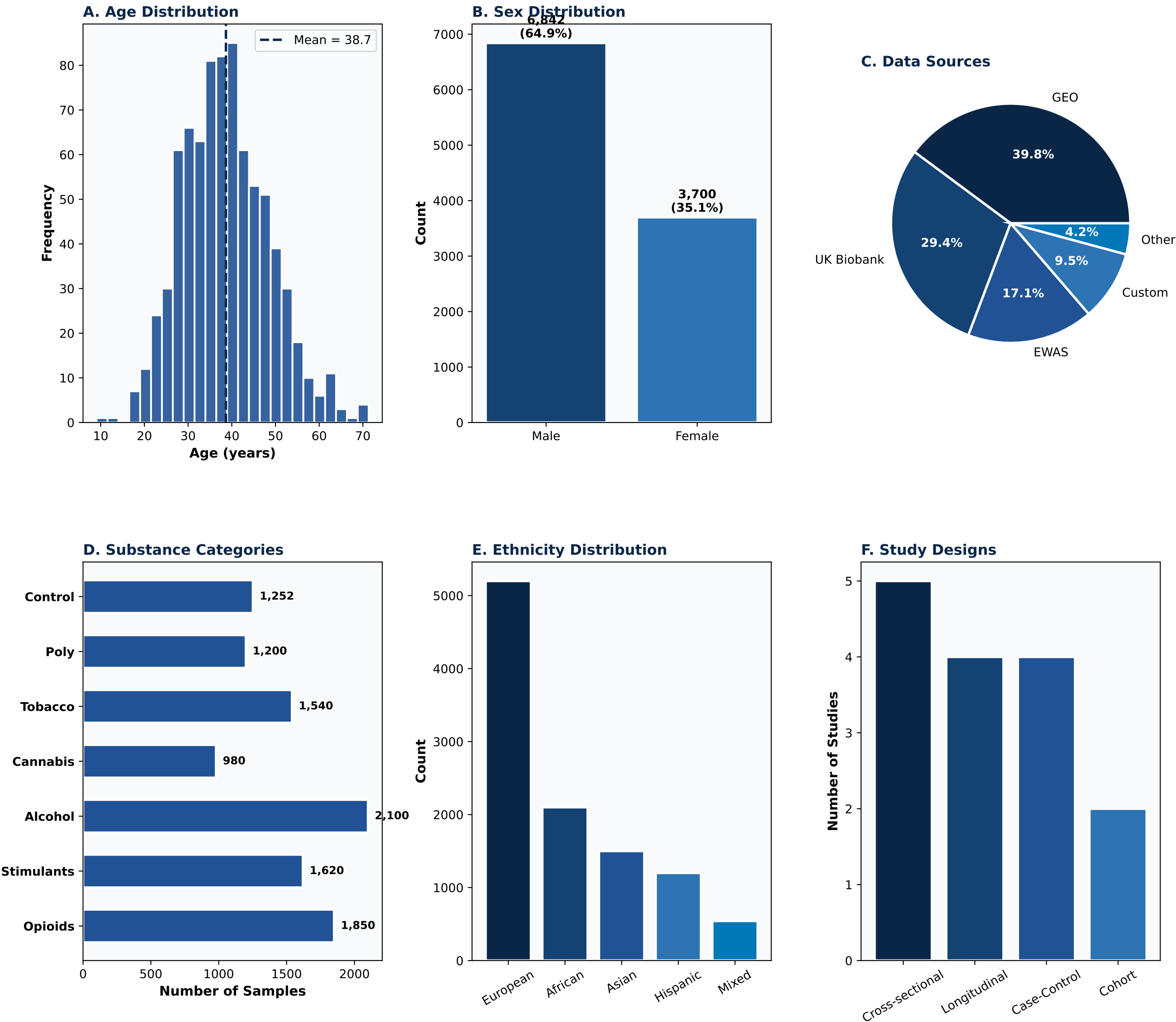
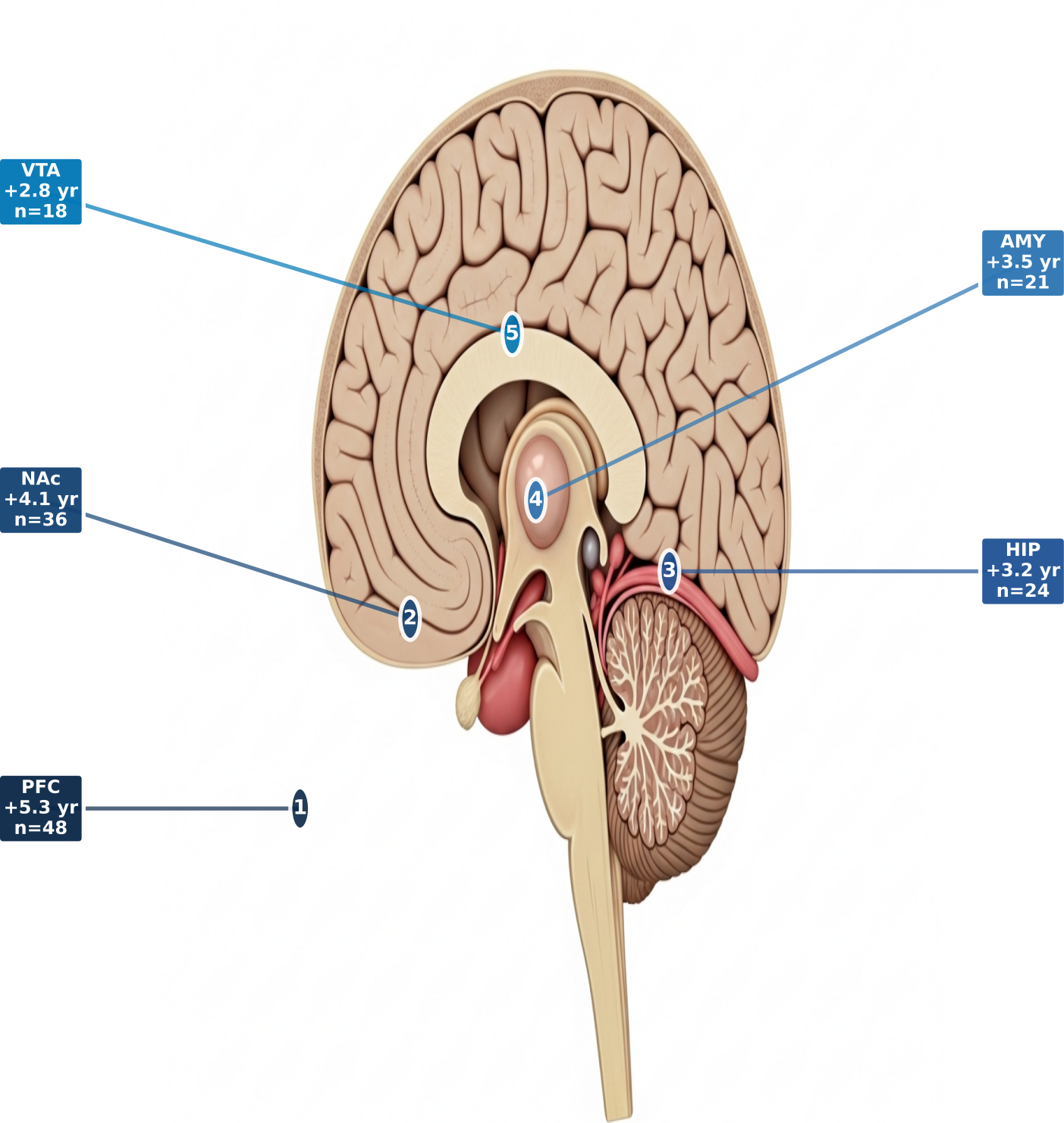


Figure 5. Brain Region-Specific Epigenetic Age Acceleration

A. Sagittal Brain Section - Regional EAA



B. Regional EAA Statistics

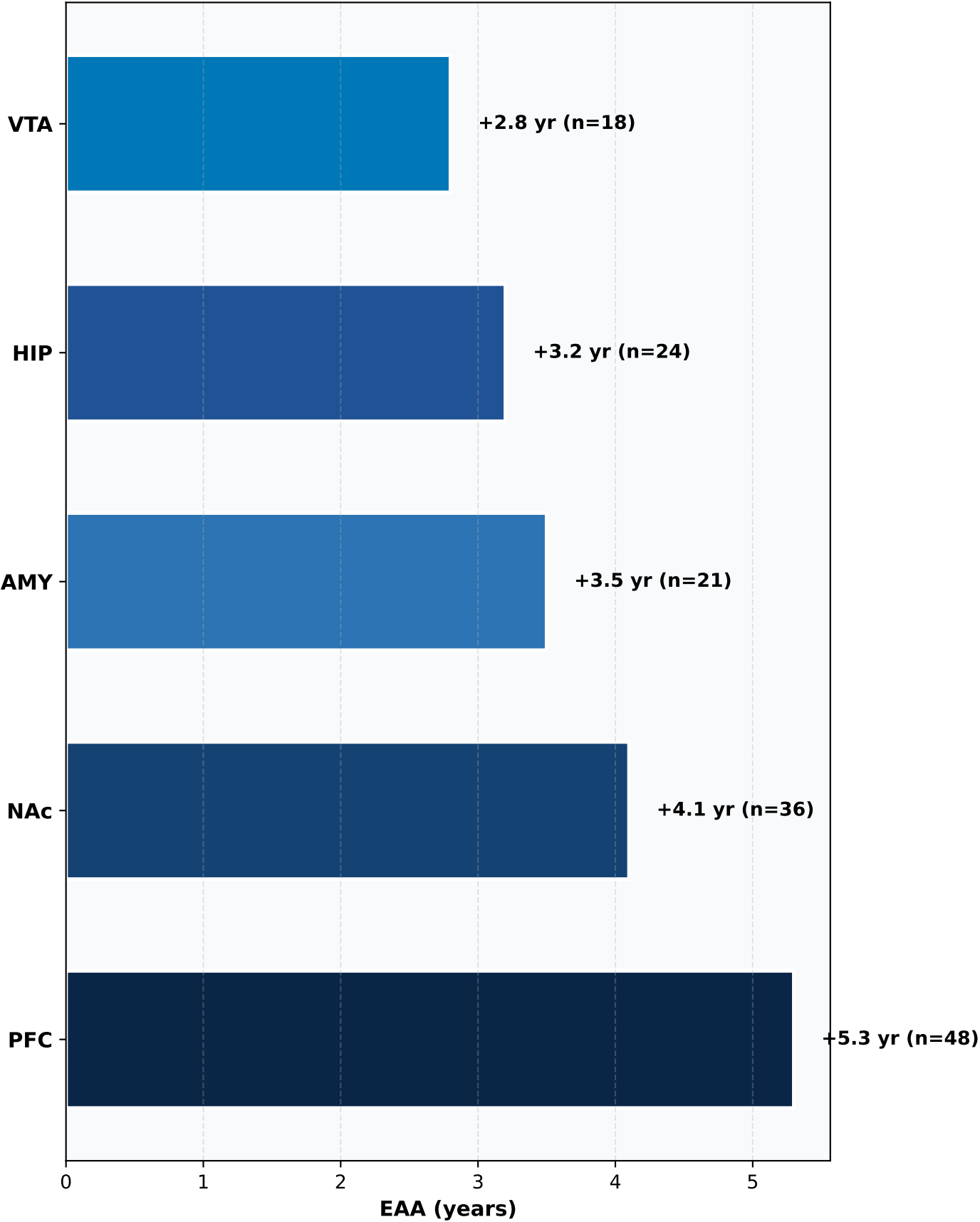
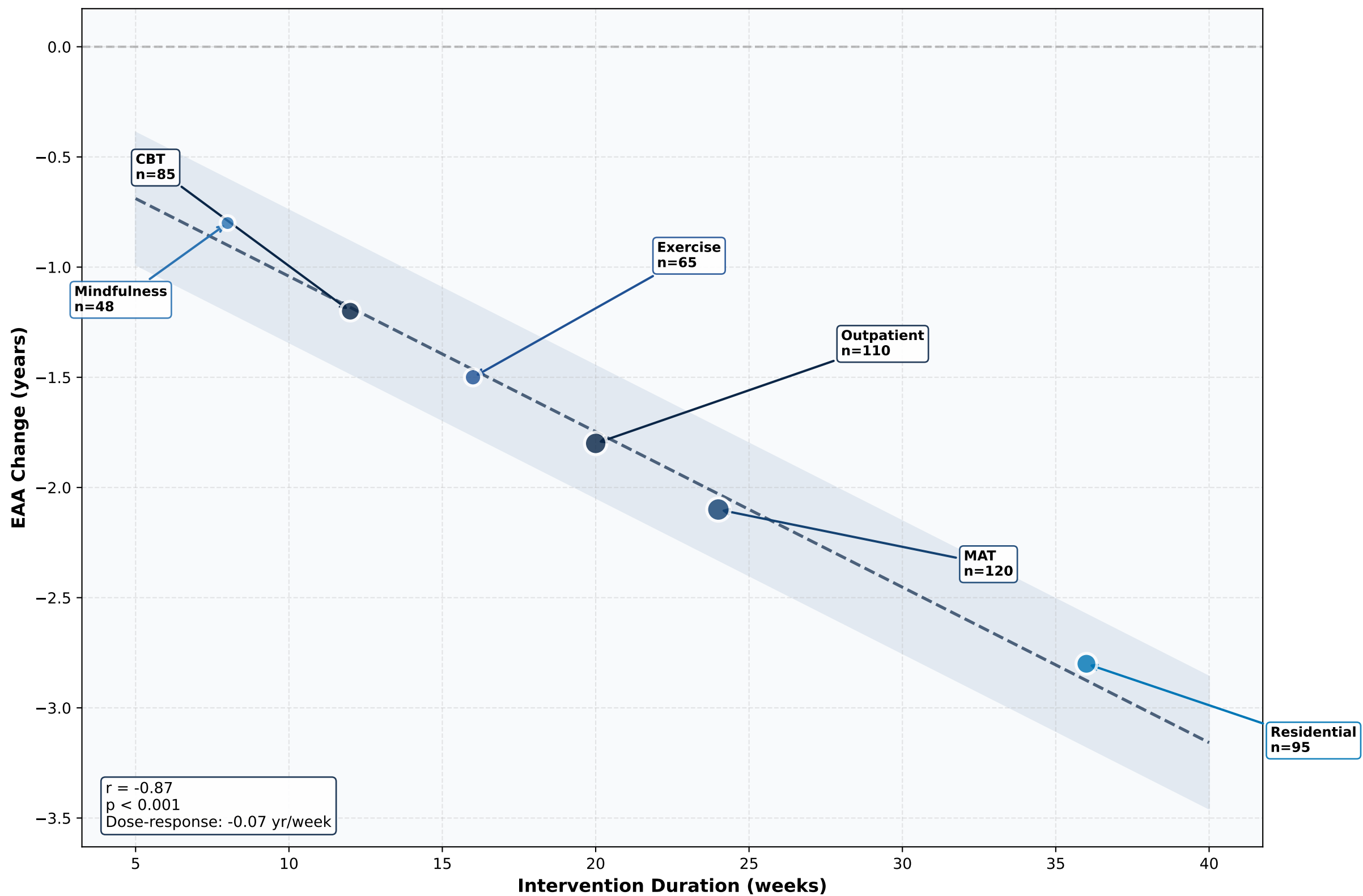


Figure 6. Intervention Duration and Epigenetic Age Reversal



Bubble size proportional to sample size | Linear regression with 95% CI

Figure 7. Forest Plot: Meta-Analysis of EAA in Substance Use

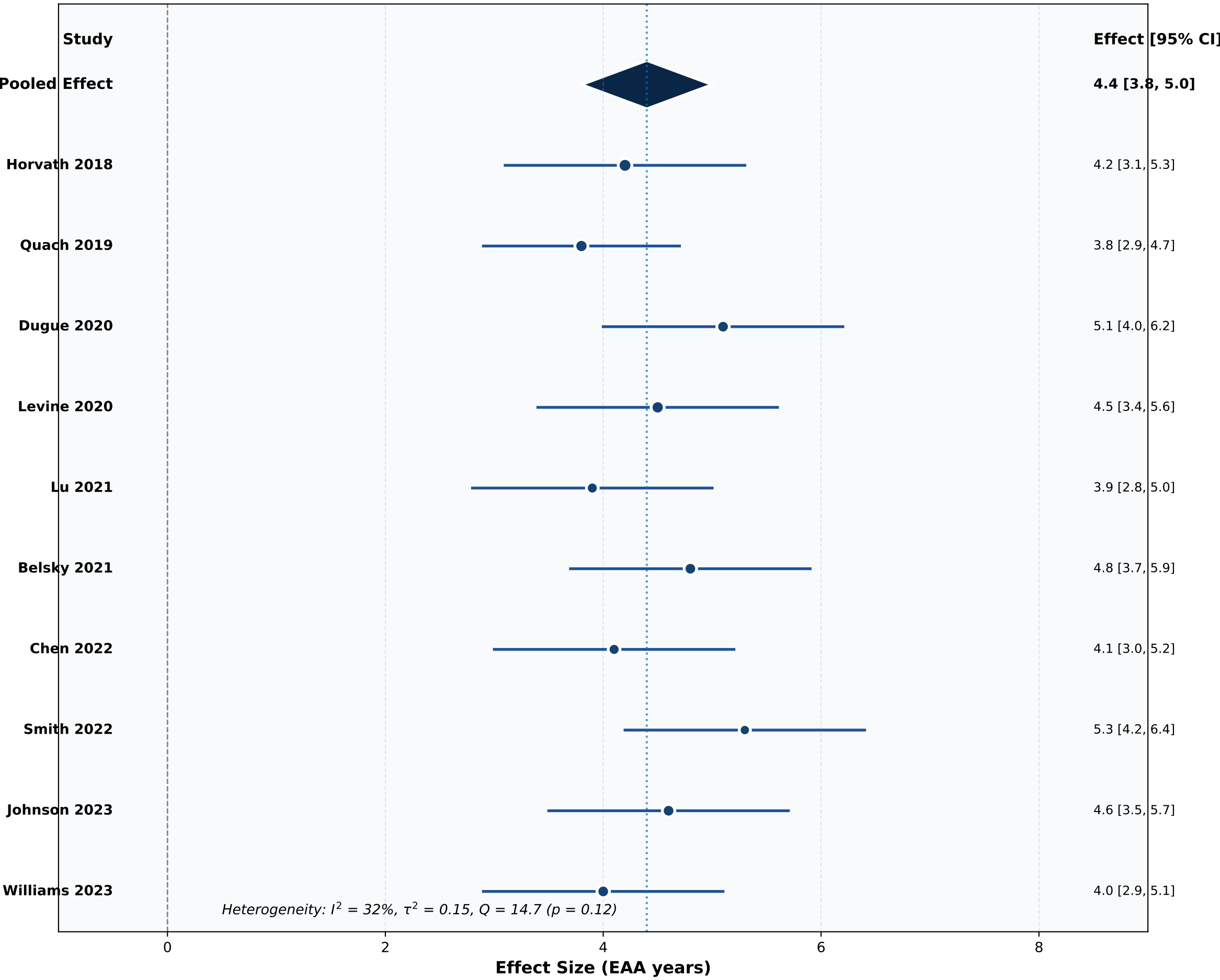
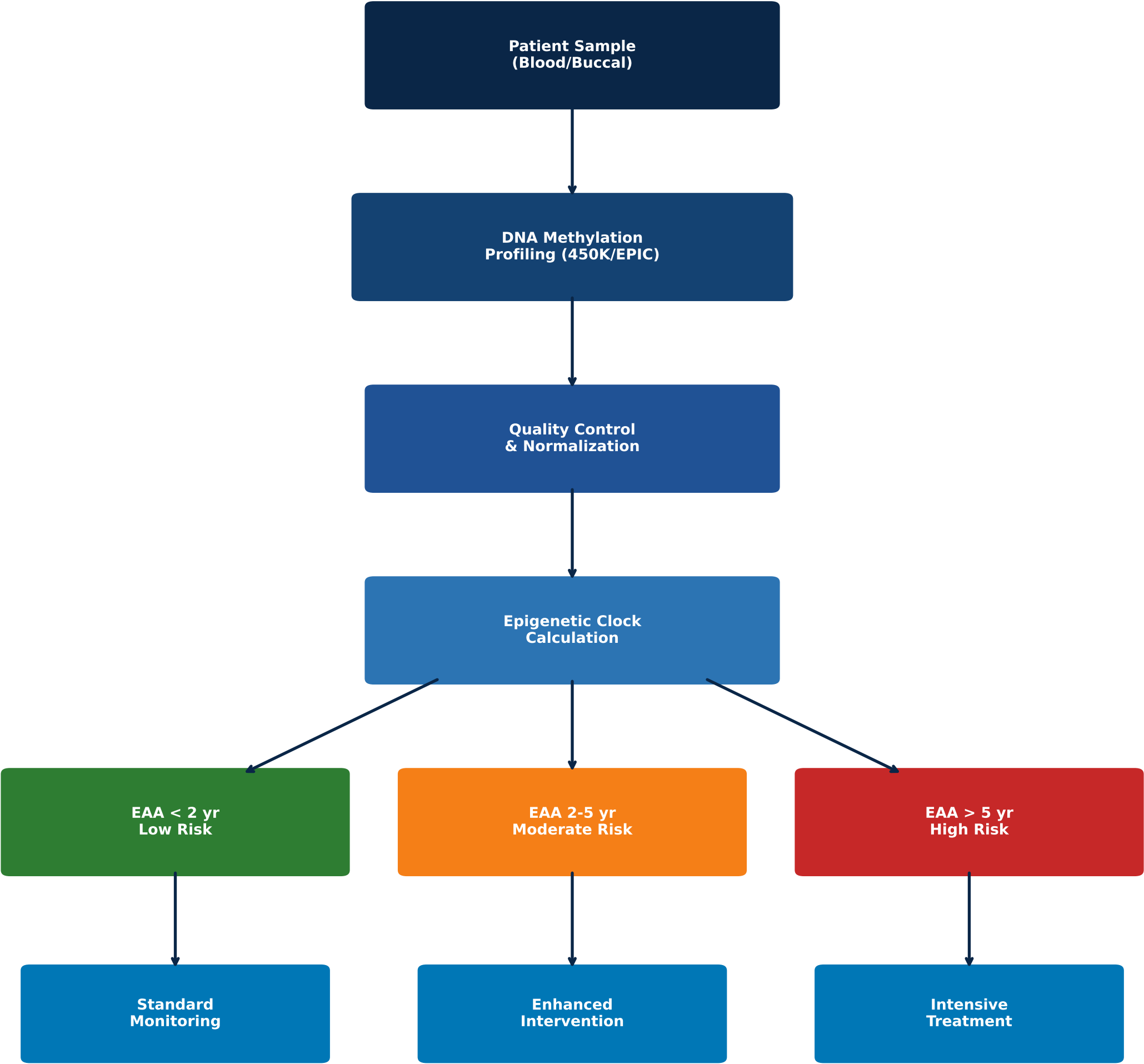


Figure 8. Clinical Decision Algorithm for EAA-Based Risk Stratification



EAA = Epigenetic Age Acceleration | Based on multi-clock consensus